

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 2.3 EXCRETION IN PLANTS: FINAL EXPLANATION

Student Assessment Document

Student Name

Class

Date

INSTRUCTIONS FOR STUDENTS

Our driving question is: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Use what you observed on the school nature walks (cut Acacia and Euphorbia, droplets on grass and arrowroot leaves, falling yellow leaves), the guttation experiment, the leaf microscope work, your class tables and your community interview portfolio. Answer every section in your own words in the blank space under each prompt. Use Claim-Evidence-Reasoning (CER) where asked, name the evidence you collected, and write in full sentences. You may add labelled sketches.

Part 1: Why plants do not need kidneys

In Lesson 1 we debated the statement 'Plants do not need kidneys.' (a) Define excretion. (b) Give three reasons why plants can manage without complex excretory organs. (c) Write a short CER statement that answers: why are plants able to live without kidneys, bladders or sweating systems?

Write your answer here:

Part 2: Routes by which wastes leave the plant

Think about the nature walk, the potted seedling under the bell jar and the plastic bag over a branch. (a) Name the routes or processes by which plants lose wastes that we studied, and say which kind of waste leaves by each. (b) Explain how guttation differs from transpiration, using the droplets on arrowroot and grass leaves on the humid morning and what you saw at the plastic bag. (c) Which of our observations on the walk showed exudation, and which showed tissue fall?

Write your answer here:

Part 3: Where insoluble wastes go

In Lesson 3 you collected young green leaves and old yellow or brown leaves and examined them under the light microscope. (a) What wastes can a plant not dissolve and remove in liquid form, and where does it deposit them? (b) Explain, using a CER statement and the observations you recorded in your own microscope work, how shedding old leaves is a form of excretion. (c) How does this idea also help explain the resin or gum on cut Acacia and Euphorbia?

Write your answer here:

Part 4: Waste to the plant, wealth to people

Use your class table from Lesson 4 and your community interview portfolio and group poster. (a) Name the types of plant excretory products that we categorised, giving one plant source and one use for each. (b) Why can something that is waste to the plant be valuable in Kenyan markets? Give examples of places or products from our lessons. (c) Some alkaloids are harmful when abused. Describe the risk of such abuse and how people should handle these products.

Write your answer here:

Part 5: Final Explanation

Answer the driving question in your own words as a complete Claim-Evidence-Reasoning explanation: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Refer to the Acacia and Euphorbia trees, the droplets on arrowroot and grass leaves, and the falling old leaves. Then state the new question that our class now wants to investigate and say why it matters.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why plants lack excretory organs	Gives all three reasons (low metabolic rate, recycling of gases, storage of insoluble wastes) with specific evidence from the chart and mesophyll diagrams.	Gives two or three reasons with some evidence.	Gives one reason or lists reasons without evidence.
Describing the routes of waste removal	Correctly names diffusion, transpiration, guttation, exudation and tissue fall, matches each to the waste type, and clearly distinguishes guttation from transpiration using local observations.	Names most routes correctly with minor confusion between guttation and transpiration.	Names few routes or confuses them.
Use of evidence and CER reasoning	Claim is clear, evidence comes from lesson observations (Acacia, Euphorbia, arrowroot and grass droplets, leaf microscope work), and reasoning links	Claim and evidence present; reasoning is partial.	Claim is unsupported or evidence is not from the lessons.

	the evidence to the claim.		
Economic value and community uses	Accurately links products, sources and uses (gum arabic, latex, tannins, quinine, caffeine) to Kenyan markets, and explains the risk of harmful alkaloids.	Gives several correct products and uses, with limited explanation of value or risks.	Gives few or inaccurate examples.
Final explanation and new question	Integrates how wastes are removed and why they are valuable in one coherent answer and states the sustainable harvesting question with a reason.	Answer covers both parts but is not fully connected, and the new question is stated.	Answer covers only one part or omits the new question.

ATTRIBUTION AND LICENSING

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